Queestion 1



31/8/2019

Show that $(\alpha T)^* = \alpha^{-1} T^*$

Solution:

$$< x, (\alpha T^*)^* y > = < (\alpha T)x, y >$$
$$< x, (\alpha T^*)^* y > = < \alpha(Tx), y >$$
$$< x, (\alpha T^*)^* y > = \bar{\alpha} < Tx, y >$$
$$< x, (\alpha T^*)^* y > = \bar{\alpha} < x, T^* y >$$
$$< x, (\alpha T^*)^* y > = < x, \bar{\alpha} T^* y >$$
$$(\alpha T^*)^* = \bar{\alpha} T^*$$

Scanned with CamScanner

Question 2



Prove that $(S + T)^* = S^* + T^*$

Prove:

$$< x, (S + T)^* y > = < (S + T)x, y >$$
$$< x, (S + T)^* y \geq < (Sx + Tx), y >$$
$$< x, (S + T)^* y > = < Sx, y > + < Tx, y >$$
$$< x, (S + T)^* y > = < x, S^* y > + < x, T^* y >$$
$$< x, (S + T)^* y > = < x, (S^* y + T^* y) >$$
$$< x, (S + T)^* y > = < x, (S^* + T^*) y >$$

Hence we have proved that

$$(S + T)^* = S^* + T^*$$

Scanned with CamScanner

Question 3

31/8/2019

Find matrix multiplication if $(\xi_1, \xi_2, \xi_3) \leftrightarrow (\xi_1, \xi_2, -\xi_1 - \xi_2)$

Solution:

$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \begin{bmatrix} \xi_1 \\ \xi_2 \\ \xi_3 \end{bmatrix} = \begin{bmatrix} \xi_1 \\ \xi_2 \\ -\xi_1, -\xi_2 \end{bmatrix}$$

Hence required matrix is

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & -1 & 0 \end{bmatrix}$$

Scanned with CamScanner

Question 4

$$\|x_1 + x_2 + \ldots\ldots + x_n\|^2 = \|x_1\|^2 + \|x_2\|^2 + \ldots\ldots + \|x_n\|^2$$

Prove:

$$\left\|\sum_{i=1}^{n} x_i\right\|^2 = \left\langle \left\|\sum_{i=1}^{n} x_i, \sum_{i=1}^{n} x_i\right\| \right\rangle$$

$$= < x_1 + \ldots\ldots + x_n, x_1 + \ldots\ldots + x_n >$$

$$= < x_1, x_1 + \ldots\ldots + x_n > + \ldots + < x_n, x_1 + \ldots\ldots + x_n >$$

$$= \sum_{i=1}^{n} \sum_{j=1}^{n} < x_i, x_j >$$

$< x_i, x_j > = \|x_i\|^2$, if $i \neq j$ $< x_i, x_j > = 0$ and for $i = j$ then $< x_i, x_j > = \|x_i\|^2$

$$\left\|\sum_{i=1}^{n} x_i\right\|^2 = \sum_{i=1}^{n} \|x_i\|^2$$

Scanned with CamScanner

Question 5

Prove that $(ST)^* = S^*T^*$

Prove:

$$< x, (ST)^*y >=< (ST)x, y >$$
$$< x, (ST)^*y \geq< (SxTx), y >$$
$$< x, (ST)^*y >=< Sx, y >< Tx, y >$$
$$< x, (ST)^*y >=< x, S^*y >< x, T^*y >$$
$$< x, (ST)^*y >=< x, (S^*+T^*y) >$$
$$< x, (ST)^*y >=< x, (S^*T^*)y >$$

5 marks
2022

Hence we have proved that

$$(ST)^* = S^*T^*$$

Scanned with CamScanner

Question 6

Show that $\overline{N(T)} < N(T)$

Prove:

$$x \in N(T)$$
$$\exists\ (x_n) such\ that\ x_n \in N(T) + x_n \to x$$

Therefore

$$Tx_n \to Tx$$
$$x_n \in N(T)$$
$$Tx_n = 0$$
$$T_n = 0$$
$$x \in N(T)$$
$$\overline{N(T)} < N(T)$$
$$\overline{N(T)} = N(T)$$
$$N(T) is\ closed.$$

Scanned with CamScanner

Question 7

31/8/2019

Prove that $< x, ay + bz >= \bar{a} < x, y > + \bar{b} < x, z >$

Prove:

$$< x, ay + bz > = < \overline{ay + bz, x} >$$
$$< x, ay + bz > = < \overline{ay, x} > + < \overline{bz, x} >$$
$$< x, ay + bz > = \bar{a} < \overline{y, x} > + \bar{b} < \overline{z, x} >$$
$$< x, ay + bz > = \bar{a} < x, y > + \bar{b} < x, z >$$

Hence proved.

Scanned with CamScanner

Question 8

Let H_1 **and** H_2 **be Hilbert spaces,** $S: H_1 \to H_2$ **and** $T: H_1 \to H_2$ **be bounded linear operator and** α **any scalar. Then show that** $(T^*)^* = T$.

Prove:

$$< x, (T^*)^* y \geq < xT^*, y >$$
$$< x, (T^*)^* y \geq < x, Ty >$$

5 marks
2022

It implies that

$(T^*)^* = T$ **so its proved.**

Scanned with CamScanner

Question 9

Prove that the product of two bounded self-adjoint linear operators S and T on a Hilbert space H is self-adjoint if and only if the operators commute, $ST = TS$.

Prove:



$$< x, (ST)^{*}y >=< (ST)x, y >$$
$$< x, (ST)^{*}y \geq < (SxTx), y >$$
$$< x, (ST)^{*}y >=< Sx, y >< Tx, y >$$
$$< x, (ST)^{*}y >=< x, S^{*}y >< x, T^{*}y >$$
$$< x, (ST)^{*}y >=< x, (S^{*}+T^{*}y) >$$
$$< x, (ST)^{*}y >=< x, (S^{*}T^{*})y >$$

Hence we have proved that

$$(ST)^{*} = S^{*}T^{*}$$

$$(ST)^{*} = T^{*}S^{*} = TS.$$

Hence

$$ST = (ST)^{*} \iff ST = TS.$$

Scanned with CamScanner

Question 10

Let H_1 and H_2 be Hilbert spaces, $S: H_1 \to H_2$ and $T: H_1 \to H_2$ be bounded linear operator and α any scalar. Then show that $\|T^{*}T\| = \|TT^{*}\| = \|T\|^2$.

Prove:

$$\|Tx\|^2 = \langle Tx, Tx \rangle = \langle T^{*}Tx, x \rangle \leq \|T^{*}Tx\| \|x\| \leq \|T^{*}T\| \|x\|^2.$$

Taking the supremum over all x of norm 1, we obtain $\|T\|^2 \leq \|T^{*}T\|$.

we have

$$\|T\|^2 \leq \|T^{*}T\| \leq \|T^{*}\| \|T\| = \|T\|^2.$$

Hence $\|T^{*}T\| = \|T\|^2$. Replacing T by T^{*} and we also hav

$$\|T^{**}T^{*}\| = \|T^{*}\|^2 = \|T\|^2.$$

Here $T^{**} = T$

Scanned with CamScanner

Question 11

Let the operators $U: H \to H$ and $V: H \to H$ be unitary; here, H is a Hilbert space. Then show that U is isometric; thus $\|Ux\| = \|x\|$ for all $x \in H$.

Prove:

$$\| U_x \|^2 = < U_x, U_x >$$
$$\| U_x \|^2 = < x, U^* U_x >$$
$$\| U_x \|^2 = < x, Ix >$$
$$\| U_x \|^2 = < x, x >$$
$$\| U_x \|^2 = \| x \|^2$$

It proved.

Scanned with CamScanner

Question 12

Show that $< T^* y, x > = < y, Tx >$

Prove:

$$< T^* y, x > = < \overline{y, T^* x} >$$
$$< T^* y, x > = < \overline{Tx, y} >$$
$$< T^* y, x > = < y, Tx >$$

Hence its proved.

Scanned with CamScanner

Question 13

Let X and Y be inner product spaces and $Q: X \to Y$ a bounded linear operator. Then Show that $Q=0$ if and only if $\langle Qx, y\rangle = 0$ for all $x \in X$ and $y \in Y$.

Prove:

$$Q = 0 \Longrightarrow Qx = 0 \text{ for all } x \epsilon X$$
$$< Qx, y > = < 0, y >$$
$$< Qx, y > = < 0.w, y >$$
$$< Qx, y > = \ 0 < w, y >$$
$$< Qx, y > = \ 0. \ \text{ for all } x \epsilon X$$

It implies that Qx=0 and by definition Q=0. Its proved.

Scanned with CamScanner

Question 14

Let the operators $U: H \to H$ and $V: H \to H$ be unitary; here, H is a Hilbert space. Then show that $\|U\| = 1$, provided $H \neq \{0\}$.

OR

Let the operators $U: H \to H$ and $V: H \to H$ be unitary; here, H is a Hilbert space. Then show that $U^{-1} (= U^*)$ is unitary.

Prove:

Since U is bijective it shows that U^{-1} is bijective

$$(U^{-1})^* = (U^*)^* = U = (U^{-1})^{-1}.$$

Hence it implies that U^{-1} is unitary.

Scanned with CamScanner

Question 15

Let the operators $U: H \to H$ and $V: H \to H$ be unitary; here, H is a Hilbert space. Then show that U is normal.

5 marks
2022

Prove:

$$UU^* = U^*U$$

Since we know that

$$U^* = U^{-1}$$

So

$$UU^{-1} = U^{-1}U$$

$(UV)^* = (UV)^{-1} = I$ **Hence its proved that U is normal.**

Scanned with CamScanner

Question 16

5 marks
2022

In an inner product space V and x, y in V if $x \perp y$ then

$$\|x+y\|^2 = \|x\|^2 + \|y\|^2$$

Prove:

$$\|x+y\|^2 = < x+y, x+y >$$

$$= < x,x > + < x,y > + < y,x > + < y,y >$$

As x and y are perpendicular so $< x,y >= 0, < y,x >= 0$

$$\|x+y\|^2 = < x,x > + < y,y > = \|x\|^2 + \|y\|^2$$

Education Information Expert
Prepared by Mubashar Ali

Scanned with CamScanner

Question 17

$$\|x+y\|^2+\|x-y\|^2=2\left(\|x\|^2+\|y\|^2\right) \quad \text{for all } x,y \in V$$

Prove:

$$\|x+y\|^2 = < x+y, x+y >$$
$$= < x,x > + < x,y > + \overline{< x,y >} + < y,y >$$
$$= \|x\|^2 + 2\,\mathrm{Re} < x,y > + \|y\|^2 \quad \text{....(i}$$

Replace y=-y

$$\|x-y\|^2 = < x+y, x+y >$$
$$= < x,x > - < x,y > - \overline{< x,y >} + < y,y >$$
$$= \|x\|^2 - 2\,\mathrm{Re} < x,y > + \|y\|^2 \quad \text{.....(ii}$$

Adding (i and (ii

$$\|x+y\|^2+\|x-y\|^2=2\|x\|^2+2\|y\|^2$$

That we have to prove.

Education Information Expert
Prepared by Mubashar Ali

Scanned with CamScanner

Question18

For any two elements x,y is an inner product space V,

$$|< x,y >| \le \|x\|.\|y\|$$

Prove:

Education Information Expert
Prepared by Mubashar Ali

If x=y=0 then 0=0

Let at least one of x and y is not equal to zero

Let $|< x+\lambda y, x+\lambda y >| \ge 0$ by definition

$$< x, x+\lambda y > + < \lambda y, x+\lambda y >$$
$$< x, x+\lambda y > + y < y, x+\lambda y >$$

Scanned with CamScanner

Question 19

SHOW THAT T is uniquely determinal if the image $y_i = Te_i$ of n basis vectors $e_1, \ldots\ldots, e_n$ are prescribed.

5 marks
2022

Prove:

$$y=Tx \;;\; y \in Y \quad \{b_1, \ldots\ldots, b_n\}$$

$$y=\eta_1 b_1 + \eta_2 b_2 + \ldots\ldots\ldots + \eta_r b_r$$

$$Te_i \in Y, \qquad Te_i = \tau_{i1} b_1 + \tau_{i2} b_2 + \ldots\ldots\ldots + \tau_{ir} b_r$$

$$Te_i = \sum_{j=1}^{r} \tau_{ij} b_j$$

$$y = \sum_{j=1}^{r} \eta_j b_j = \sum_{i=1}^{n} \xi_i Te_i = \sum_{i=1}^{n} \xi_i \sum_{j=1}^{r} \tau_{ij} b_j$$

Comibinig these two summation

$$y = \sum_{j=1}^{r} \left(\sum_{i=1}^{n} \tau_{ij} \xi_i \right) b_j$$

$$\eta_j = \sum_{i=1}^{n} \tau_{ij} \xi_i$$

The image $y=Tx=\sum \eta_j b_j$ of $x = \sum \xi_i Te_i$ can be obtained from

$$\eta_j = \sum_{i=1}^{n} \tau_{ij} \xi_i$$

Scanned with CamScanner

Question 20

3 marks
2022

Sum and Scalar multipication of Dual.

Prove:

as follows. The SUM $f_1 + f_2$ of two functionals f_1 and f_2 is the functional s whose value at every $x \in X$ is

$$s(x) = (f_1 + f_2)(x) = f_1(x) + f_2(x);$$

the Product αf of a scalar α and a functional f is the functional p whose value at $x \in X$ is

$$p(x) = (\alpha f)(x) = \alpha f(x).$$

Scanned with CamScanner

Question 21

Define Inner Product Space:

The pair $(V, \langle \bullet, \bullet \rangle)$ is called an inner product space.

Define Appolonius Identity:

$$\|z-x\|^2+\|z-y\|^2=\frac{1}{2}\|x-y\|^2+2\left\|z-\frac{1}{2}(x+y)\right\|^2 \quad , \quad x,y,z \in V$$

Define Equal Operators:

Let T_1 and T_2 be two operators defined on equal i.e. $T_1 = T_2$ if they have the same domain i.e. $D(T_1) = D(T_2)$

Scanned with CamScanner

Question 22

2 marks 2022

Define Orthonormal System.

Orthonormal systems are those functions which are real or complex whose inner product with itself results in 1 and with other functions result in 0.

Define Polarization Identity.

2 marks 2022

For any x, y in complex inner product space

$$\langle x,y\rangle=\frac{1}{4}\left\{\|x+y\|^2-\|x-y\|^2+i\|x+iy\|^2-i\|x-iy\|^2\right\}$$

We have to prove this complex inner product space.

Scanned with CamScanner

Question 23

2 marks 2022

Define Bounded Linear Function.

Let X and Y be normed spaces and $T: D(T) \to Y$ a linear operator, where $D(T) \subset X$. Th operator T is said to be bounded if there is a real number c such that for all $x \in D(T)$.

$\|Tx\| \leq c\|x\|$

Define Canonical Mapping. 2 marks 2022

$C: X \to X^{**}$ this mapping is called canonical mapping of X into X^{**} defined as $x \mapsto g_x$.

Scanned with CamScanner

Question 24

2 marks 2022

Define Unitary Operator.

Abounded Linear Operator $T: H \to H$ on a Hilbert Space H is said to be Unitary Operator if T is bijective and $T^* = T^{-1}$.

2 marks 2022

Define Normal Operator.

Abounded Linear Operator $T: H \to H$ on a Hilbert Space H is said to be Unitary Operator if T is bijective and $TT^* = T^*T$

Define Extension of T.

Extension of T on a set $M \supset D(T)$ is defined as $\tilde{T}: M \to Y$ or

$$\tilde{T}x = Tx \text{ for all } x \in D(T)$$

Scanned with CamScanner

Question 25

Write examples of Bounded Linear Operators

Identity Operator, Zero Operator, Differentiation Operator, Integral Operator

31/8/2019

Define Self Adjoint Operator or Hermitian Operator.

Abounded Linear Operator $T: H \to H$ on a Hilbert Space H is said to be Self Adjoint Operator or Hermitian Operator if $T^* = T$.

Give an example of function which is not linear.

$\| . \| : x \to R$ is functional but not linear.

2 marks 2022

Find basis of R(T).

$(\xi_1, \xi_2, -\xi_1, -\xi_2) = \xi_1(1,0,-1) + \xi_2(0,1,-1)$

So the basis of R(T)= $\{(1,0,-1),(0,1,-1)\}$

Scanned with CamScanner

Question 26

2 marks 2022

Give any two examples of Linear Functional.

Space C[a, b] and Space l^2

Define Linear Functional:

A linear functional f is a linear operator with domain in a vector space X and range in the scalar field K of X; thus $f: D(f) \to K$. where K=R if X is real and K=C if X is complex.

Education Information Expert
Prepared by Mirza Nasir Ali

Define Bounded Linear Extension:

$T: D(T) \to Y$ be a bounded linear operator, where D(T) lies in a normal space X and Y is a Banach Space then T has an extension $\tilde{T}: \overline{D(T)} \to Y$. Where $\tilde{T}$ is a bounded linear operator of norm.

$$\| \tilde{T} \| = \| T \|$$

Scanned with CamScanner

Bounded Linear Operator:

Let X and Y be normed spaces and $T : D(T) \to Y$ a linear operator, where $D(T) \subset X$.

The operator T is said to be bounded if there is a real number c such that for all $x \in D(T)$.

$$\|Tx\| \leq c\|x\|$$

Let T be a bounded linear operator as defined before then an alternate formula for the norm of T is

$$\|T\| = \sup_{\substack{x \in D(T) \\ \|x\|=1}} \|Tx\|$$

2 marks

2022

The norm defined on T satisfies (N1) to (N4).

Examples of Bounded Linear Operators

Identity Operator
Zero Operator
Differentiation Operator
Integral Operator
Matrix

Linear Function:

Algebraic Dual Space
Second Algebraic Dual Space
Canonical Mapping

Algebraic Dual Space

Set of all linear function defined on a vector space X is itself a vector space and called Algebraic Dual Space and denoted by X^*

Conical Mapping:

$C : X \to X^{**}$ this mapping is called canonical mapping of X into X^{**} defined as $x \mapsto g_x$.

Linear Function (Examples):

Space $C[a\ \ b]$
Space l^2

Linear Operators and functional on finite Dimensional Spaces:

Finite dimensions mean basis which have finite many elements.

Let X and Y bef.gfinite dimension vector spaces over the same field.

Let $T : X \to Y$ be a linear operator. let $E = \{e_1, \ldots\ldots\ldots, e_n\}$ be the basis for X and $B = \{b_1, \ldots\ldots\ldots, b_n\}$ be the basis for Y.

$x \in X$, $\quad x = \xi_1 e_1 + \xi_2 e_2 + \ldots\ldots\ldots\ldots + \xi_n e_n$

$$y = Tx = T\left(\sum_{k=1}^{n} \xi_k e_k\right) = \sum_{k=1}^{n} T(\xi_k e_k) = \sum_{k=1}^{n} \xi_k T(e_k)$$

Scanned with CamScanner

for all $x, y \in V$, $a \in F$

$$< x, ay > = \overline{< ay, x >} = \overline{a < y, x >}$$

$$= \bar{a} \overline{< y, x >} = \bar{a} < x, y >$$

Theorem:

For any two elements x, y is an inner product space V,

$|< x, y >| \leq \|x\| . \|y\|$, the define norm is $\|x\| = \sqrt{< x, x >}$, $x, y \in V$

Proof:

If x=y=0 then 0=0

Let at least one of x and y is not equal to zero

Let $|< x + \lambda y, x + \lambda y >| \geq 0$ by definition

$$< x, x + \lambda y > + < \lambda y, x + \lambda y >$$

$$< x, x + \lambda y > + y < y, x + \lambda y >$$

Theorem:

In an inner product space V, the function $\| . \| : V \to \mathbb{R}^+$ given by

$\|x\| = \sqrt{< x, y >}$, $x \in V$ defines a norm in V.

2 marks 2022

Proof:

N1: $\|x\| \geq 0$

For a $x \in V$, $\|x\| = \sqrt{< x, x >} \geq 0$ as $< x, x > \geq 0$

N2:

$\|x\| = 0 \Leftrightarrow \sqrt{< x, x >} = 0 \Leftrightarrow < x, x > = 0 \Leftrightarrow x = 0$

N3: $\|\alpha x\| = |\alpha| \|x\|$

now $\|\alpha x\| = \sqrt{< \alpha x, \alpha x >} \Rightarrow \|\alpha x\|^2 = < \alpha x, \alpha x >$

$\Rightarrow \|\alpha x\|^2 = \alpha \bar{\alpha} < x, x > = |\alpha|^2 \|x\|^2$

N4: $\|x + y\| \leq \|x\| + \|y\|$ $\forall$ x, y $\in$ V

Scanned with CamScanner

Any sequence $\{x_n\}$ of non-zero mutually orthogonal vectors in an inner product space V is linearly independent.

Proof: do it yourself

3 marks
2022

Let $x = (x_1, x_2, \ldots\ldots x_n)$ be the orthogonal sequence.

Remark:

If $< x, x_1 >= 0$, $\forall$ i=1,2,.....,n $\Rightarrow$ $\left\langle \sum_{i=0}^{n} a_i x_i, x \right\rangle = 0$

$$\left\langle \sum_{i=0}^{n} a_i x_i, x \right\rangle = \langle a_1 x_1 + a_2 x_2 + \ldots\ldots + a_n x_n, x \rangle = a_1 \langle x_1, x \rangle + \ldots\ldots\ldots\ldots + a_n \langle x_n, x \rangle = 0$$

Scanned with CamScanner

31/8/2019

IMPORTANT SUBJECTIVE FOR FINAL TERM

Q.No.01: Prove that $< x, ay + bz >= \bar{a} < x, y > + \bar{b} < x, z >$

Proof:

$$< x, ay + bz > = < \overline{ay + bz, x} >$$
$$< x, ay + bz > = < \overline{ay, x} > + < \overline{bz, x} >$$
$$< x, ay + bz > = \bar{a} < \overline{y, x} > + \bar{b} < \overline{z, x} >$$
$$< x, ay + bz > = \bar{a} < x, y > + \bar{b} < x, z >$$

Hence proved.

Q.No.02: Prove that UV is unitary OR $(UV)^* = (UV)^{-1}$

Proof:

$$(UV)^* = U^* V^*$$

Since we know that

$$U^* = U^{-1} \text{ and } V^* = V^{-1}$$

3 marks
2022

So

$$(UV)^* = U^{-1} V^{-1}$$
$$(UV)^* = (UV)^{-1}$$

It shows that UV is unitary.